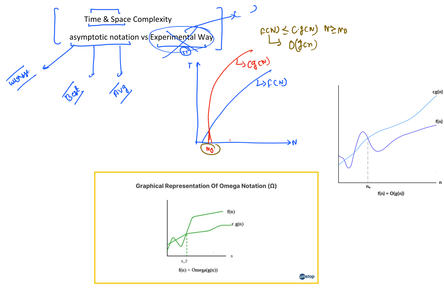
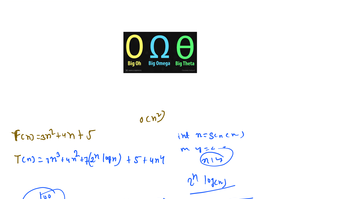
Lec-11

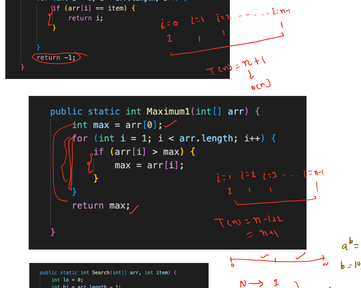




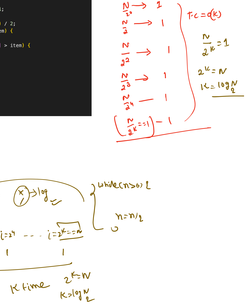




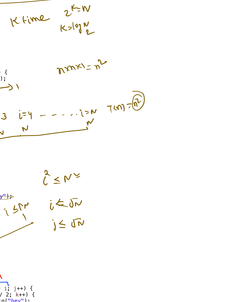





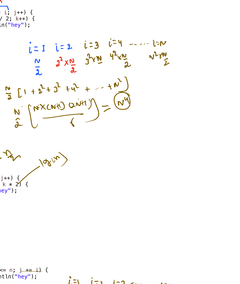

```
public static int Search(int[] arr, int item) {

    for (int i = 0; i < arr.length; i++) {
      if (arr[i] == item) {
          return i;
      }
    }
    return -1;
}
```

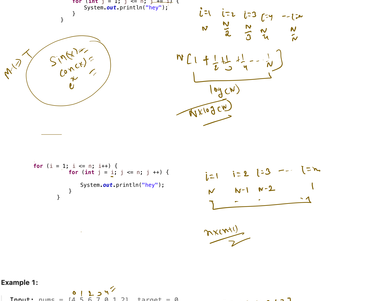


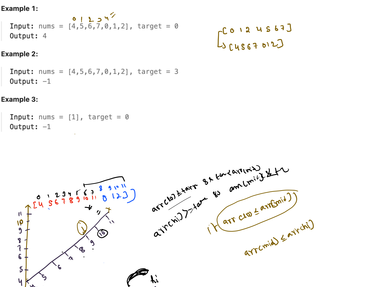

```
public static int Maximum1(int[] arr) {
    int max = arr[0];
    for (int i = 1; i < arr.length; i++) {
      if (arr[i] > max) {
          max = arr[i];
      }
    }
    return max;
}
```

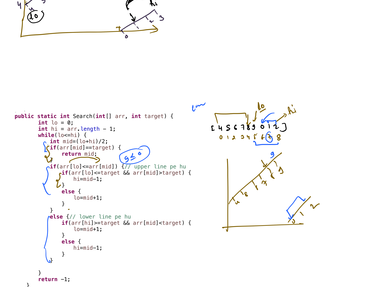



```
public static int Search(int[] arr, int item) {
    int lo = 0;
    int hi = arr.length - 1;
    while (lo <= hi) {
        int mid = (lo + hi) / 2;
        if (arr[mid] == item) {
            return mid;
        } else if (arr[mid] > item) {
            hi = mid - 1;
        } else {
            lo = mid + 1;
        }
    }
    return -1;
}
```













**Example 1:**

**Input:** nums = [4,5,6,7,0,1,2], target = 0
**Output:** 4

**Example 2:**

**Input:** nums = [4,5,6,7,0,1,2], target = 3
**Output:** -1

**Example 3:**

**Input:** nums = [1], target = 0
**Output:** -1




```
public static int Search(int[] arr, int target) {
    int lo = 0;
    int hi = arr.length - 1;
    while(lo<=hi) {
      int mid=lo+hi)/2;
      if(arr[mid]==target) {
          return mid;
      }
      if(arr[lo]<=arr[mid]) {// upper line pe hu
          if(arr[lo]<=target && arr[mid]>target) {
              hi=mid-1;
          }
          else {
              lo=mid+1;
          }
      } else {// lower line pe hu
          if(arr[hi]>=target && arr[mid]<target) {
              lo=mid+1;
          }
          else {
              hi=mid-1;
          }
      }
    return -1;
}
```